

# Aahana Bobade

ML Engineer · Data Scientist · Full Stack AI Developer

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## SUMMARY

ML Engineer & Data Scientist with 1 year of production AI/ML experience shipping LLM systems and data-driven products to 50+ enterprise users. Built GraphRAG pipelines with 35% precision gain over vanilla RAG, designed multilingual agentic chatbots that lifted completion rates by 40%, and delivered autonomous scheduling AI recovering 4,000+ agent-hours/year. Skilled in end-to-end ML: data analysis, model evaluation (RAGAs, precision/recall, F1), pipeline design, and production deployment. Computer Engineering graduate, GPA 9.28/10. 100+ LeetCode. Open to ML Engineer and Data Science roles.

## EXPERIENCE

### Software Developer · EduvanceAI

Aug 2025 - Present | Mumbai

- Built a production RAG-based Sales Copilot for a 50-agent field team: applied TSP route optimization + LLM dealer-scoring to auto-generate 30-day visit plans, cutting manual planning from ~2 hrs/week to under 25 min per agent, recovering an estimated 4,000 agent-hours annually and reducing coordination errors to zero.
- Engineered multilingual agentic simulation chatbots (LangChain + LangGraph) for a GenAI-powered LMS across 5 languages; contextual memory and dynamic scenario branching drove a 40% lift in module completion rate, eliminating human facilitators per training session and reducing L&D overhead.
- Designed and ran a structured ML evaluation study across 4 GenAI frameworks, measuring retrieval precision, hallucination rate, and reasoning quality using RAGAs and LLM-as-judge; findings adopted as the org-wide LLM evaluation standard, reducing model iteration cycles by ~30%.

### UX Designer · Zepto Digital Labs

Jun - Aug 2023 | Thane

- Redesigned simulation platform UI in Figma; delivered 12 high-fidelity prototypes validated with 20+ users via structured usability tests, reduced key task completion time by 30% and cut UI-related support tickets post-launch.

### Backend Intern · Laser Technologies Pvt. Ltd.

Jun - Jul 2023 | Navi Mumbai

- Cut average DB query latency by 20% across 3 enterprise web apps by rewriting N+1 queries and adding composite indexes; triaged and resolved 10+ P1 incidents with zero data loss or SLA breach.

## PROJECTS

### Little Angel Foundation - Full-Stack Platform · React · Node.js · PostgreSQL · NeonDB · Vercel 2026

- Deployed a full-stack NGO platform with CMS and volunteer coordination portal; reduced content update turnaround from 3 days to under 10 minutes for non-technical staff, zero reported downtime since launch with 5 concurrent admin roles.

### OrgMind - Hybrid GraphRAG System · Neo4j · Pinecone · GPT-4 · LangChain · RAGAs · Python 2026

- Designed and benchmarked a GraphRAG pipeline combining Neo4j knowledge-graph traversal with Pinecone vector search; evaluated across 500+ multi-hop organizational queries using RAGAs, achieved context retrieval precision of 0.71 vs. 0.53 baseline (vanilla RAG), a 35% relative gain, and reduced hallucination rate on relational queries by 28%.
- Engineered a 4-stage inference pipeline (query decomposition → graph retrieval → vector retrieval → response synthesis); systematic ablation study showed graph-retrieval stage alone accounted for 60% of precision improvement, informing architecture decisions for production deployment.

### Smart Resource Tracker - LRU Cache + Analytics Dashboard · FastAPI · React · Python · Locust 2026

- Implemented a thread-safe LRU cache (doubly-linked list + hashmap) with TTL expiry; load-tested with Locust at 500 concurrent users, delivered 3x throughput over naive dict-based implementation with p99 latency under 12ms.
- Built a real-time React analytics dashboard exposing hit/miss ratios, eviction rates, and TTL decay curves, replaced manual log analysis, reducing root-cause debugging time by 50% and enabling data-driven cache tuning.

### Gita-GPT · LLM · RAG · Hume AI · FastAPI · AWS · TTS/STT 2023 - Present

- Fused Hume AI valence/arousal emotion signals with a custom RAG pipeline over a corpus of 700+ Bhagavad Gita verses; emotion-conditioned retrieval increased avg session depth by 60% vs. keyword-matching baseline, measured via session analytics across active users.
- Evaluated retrieval quality using MRR and context precision across 200 test queries; iteratively improved chunking strategy and re-ranking, lifting MRR from 0.61 to 0.79 over 3 model iterations.

### Safe Yatra - On-Device ML Safety App · TensorFlow.js · React Native · Geolocation API

2024 – 2025

- Trained and deployed a TensorFlow.js voice emotion classifier (70% distress-detection accuracy) fully on-device with zero server inference latency, enabling offline SOS detection critical for low-connectivity scenarios.
- Analyzed false-negative failure modes from 150 test audio samples; identified that background noise accounted for 68% of misclassifications, leading to targeted data augmentation that improved accuracy by 8 percentage points.

### LeetPush - Developer Productivity Extension · JavaScript · GitHub API · Chrome Extensions API

2026

- Built a Chrome Extension intercepting window.fetch to auto-sync accepted LeetCode solutions to GitHub across 15 languages; auto-generates categorized READMEs and tracks submission streaks, zero third-party service dependency.

## TECHNICAL SKILLS

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**ATS Keywords:** Python, PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, RAG, GraphRAG, Hugging Face Transformers, RAGAs, MLOps, NLP, spaCy, NLTK, Prompt Engineering, Agentic Workflows, Model Evaluation, A/B Testing, SQL, Pandas, NumPy, Matplotlib, Tableau, FastAPI, React, Node.js, Docker, AWS, PostgreSQL, MongoDB, Neo4j, Pinecone, FAISS

**ML / AI:** PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, RAG Pipelines, GraphRAG, Hugging Face Transformers, RAGAs, Prompt Engineering, Agentic Workflows, NLP (spaCy, NLTK), Model Evaluation, A/B Testing

**Data & Analytics:** Python (Pandas, NumPy, Matplotlib, Seaborn), SQL (window functions, CTEs), Tableau, Power BI

**Languages:** Python (primary), SQL, JavaScript, TypeScript, Java

**Backend & Infra:** FastAPI, Node.js, React, REST APIs, Docker, AWS EC2, GitHub Actions, Supabase

**Databases:** PostgreSQL, MongoDB, Neo4j, Pinecone, FAISS

**Core CS:** Data Structures & Algorithms (100+ LeetCode), System Design, OOP, Operating Systems

## EDUCATION

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### B.E. Computer Engineering · SIES Graduate School of Technology, Nerul

2021 – 2025

Minor: Artificial Intelligence & Machine Learning

GPA: 9.28 / 10